

X04 SITE REMEDIATION | O (MAX : 1 PT)

Intent : Promote the safer development of potentially contaminated sites by assessing and mitigating hazards.

Summary : This WELL feature requires site assessment, testing and remediation for the development of contaminated sites.

Issue : Contaminated soil, usually associated with past industrial activities, can leach toxic chemicals into nearby groundwater or surface waters, accumulate in sediments, volatilize and pose hazards to indoor air in buildings on the premises or be carried by wind-borne dust.¹ When left unmanaged, contaminants from such sites can pose risks to those who live and work nearby through inhalation, ingestion or dermal contact.² Contamination in these sites, which are known as brownfields, ¹ may complicate development if not properly addressed.

Solutions : Site assessment and remediation can reduce risk of exposure to populations that live in proximity to contaminated sites. Cleanup of contaminated sites that can present environmental (e.g. air, water, soil) and human health hazards helps protect the public from associated hazards and encourages environmentally responsible growth,³ further preserving undeveloped land.

PART 1 ASSESS AND MITIGATE SITE HAZARDS (MAX : 1 PT)

For All Spaces:

The following requirements are met:

- a. Site was used for or affected by past or present industrial activities (e.g., hazardous waste storage, fuel station, manufacturing plant, on-site dry cleaners, automotive repair or brownfields).
- b. Site is assessed for potential contamination in soil or underground water from past uses or surrounding conditions using one of the following:
 1. Local applicable regulation for environmental site assessments.
 2. Guidelines provided in the standard ASTM E1527-05 (Phase I site assessments).
- c. If the site investigation establishes the potential presence of contaminants, the project implements a sampling strategy to quantify contamination and determine remediation needs according to local regulations or ASTM E1903-97 (Phase II site assessment) guidelines.
- d. If remediation is required, the project implements a sustainable remediation plan before, during and after construction that integrates the following:^{3,4}
 1. A risk-based approach to sustainable remediation (risk assessment/risk-benefit analysis).
 2. A tiered approach to assessment and an appraisal of remediation options.
 3. Safe working practices for workers during remediation.
 4. Record keeping of decision-making and assessment processes.
 5. Protocol for engaging stakeholders, including management of the impacts on the community.

REFERENCES

1. U. S. Environmental Protection Agency. Contaminated Land. Report on the Environment (ROE) Web site. <https://www.epa.gov/report-environment>. Published 2017. Accessed February 5th, 2020.
2. University of the West of England. Soil Contamination: Impacts on Human Health Science for Environment Policy. Bristol: European Commission;2013.
3. Rizzo E, Bardos P, Pizzol L, et al. Comparison of international approaches to sustainable remediation. Journal of Environmental Management. 2016;184:4-17.
4. International Standards Organisation. Soil quality — Sustainable remediation. In:2017.